

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**ORGANISATION OF ISLAMIC COOPERATION (OIC)****Department of Computer Science and Engineering (CSE)****SEMESTER FINAL EXAMINATION****SUMMER SEMESTER, 2020-2021****DURATION: 3 HOURS****FULL MARKS: 150****SWE 4803: Software Project Management**

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. a) Identify and explain the important characteristics of software development projects which make them harder to manage compared to other types of projects. 8
(CO1)
(PO2)
- b) Why software development houses prefer to use project organization over functional organization? How the project manager differs from functional or operational manager? 5+4
(CO1)
(PO1)
- c) General Electric (GE) has the opportunity to invest in 2 projects. Project A requires an investment of \$1000000 which will give a return of \$300000 each year for 5 years. Project B requires an investment of \$750000 which will give a return of \$100000, \$150000, \$200000, \$250000, and \$ 250000 for the next 5 years. In which project GE should invest if the discount rate is 7%? 8
(CO1)
(PO11)

2. a) **Table 1: Table for Question 2. (a)**

10+ 2+5
(CO3)
(PO2)

Project Activity	Precedence Activity	Duration (Weeks)		
		Optimistic	Most likely	Pessimistic
Start		0	0	0
C	Start	2	3	4
D	C	2	7	9
A	C	5	7	12
E	D	1	3	8
B	A, E	4	5	6
F	E	2	3	4
End	B, F	0	0	0

- i. Estimate the duration and standard deviation of each task provided in Table 1 using PERT (beta distribution).
- ii. What is the estimated total duration of the project according to PERT?
- iii. What should be the safe statement to the sponsors about the estimation of project duration?
- b) "Lag provides mandatory delay to the Successor Activity and Lead provides acceleration to the Successor Activity" – justify with proper example. 8
(CO3)
(PO2)
3. a) While creating the agenda of a common status meeting for two different projects, one of the team member mentions a very specific item that should be added in the meeting agenda. One of the project managers says that "We cannot deal with this item now!" Few other team members say that the team is not ready to resolve the item during the meeting. But after having a discussion with both the project teams, another project manager decides to put the item on the agenda as an initial discussion item. What sort of conflict resolution techniques are the project managers utilizing in the given scenario? 5
(CO3)
(PO2)

- b) CSE Department of IUT is planning to organize an App development contest on the last Friday of July 2022. You are selected as a project manager for this big event. You are allowed to start working on this project from 1st week of May 2022. Develop a project charter for your project. 15 (CO3) (PO3)
- c) A company is making an effort to improve its project performance and create historical records of past projects. What is the best way to accomplish this? 5 (CO3) (PO2)
4. a) You want to let your team know that the weekly project status report is now officially due by 4:00 PM on each Thursday. Which type of communication would you use? 5 (CO3) (PO2)
- b) A project manager for a large bank is in-charge of developing a new certificate of deposit product, which needs IT system development. His internal IT staff does not have expertise in the technology needed for this project. So, he must contract out the work. Unfortunately, he does not have the time to develop a detailed procurement statement of work, but this effort is likely to be large. In this situation, what is the best contract type the project manager should select? 8 (CO3) (PO4)
- c) Formulate with your own example how the earned value chart depicts scheduled progress, actual cost, and actual progress (earned value) to allow the determination of cost, schedule, and time variances. 12 (CO3) (PO3)
5. a) For your upcoming project, you decided to develop 80 functionalities of your product within 5 days by some third-party Software development firm. The plan for 5 days and the actual development till day 3 is given in Table 2. Also, it is estimated that development cost for each functionality will be 1000\$. And as a project manager you are performing monitoring and control activities on day 3. 6+4 (CO3) (PO2)

Table 2: Table for Question 5. (a)

	Day 1	Day 2	Day 3	Day 4	Day 5
Functionalities planned to be built	10	13	17	20	20
Functionalities actually built	8	12	16	.	
Actual cost for the day	8000	12000	16000		

- i. Based on the given information analyze the impact of SV, CV, SPI, and CPI on your project.
- ii. As the third-party organization failed to deliver the expected outcome, you decided to revise the budget of this project as 78000\$. Calculate the TCPI for this change.
- b) In one of the outsourced software development contractual agreements, the buyer and seller agreed to a cost of \$300,000 and a profit/fee of \$30,000. The buyer has informed that the ceiling price will be \$360,000. Beyond the target cost, the sharing ratio between the buyer and seller will be 60:40. What is the PTA? 8 (CO3) (PO2)
- c) Does profit/fee become zero at PTA? Justify your answer with proper example. 7 (CO3) (PO2)
6. a) As a project manager you need to decide whether to invest \$120M to build a new data center, or to invest only \$50M to upgrade the existing data center. If you decide to build a new one, then there is 60% chance that the new data center will help to generate revenue of \$200M and 40% chance of generating revenue of \$90M. Also, if you decide to upgrade the existing data center then there is 60% chance that it will help to generate revenue of \$120M and 40% chance of generating revenue of \$60M. Using proper data analysis method make your decision for the given scenario. 12 (CO4) (PO4)
- b) "Designing redundancy into a system may reduce the impact from a failure of the original component." – What kind of strategy is this for dealing with risk? 5 (CO4) (PO2)
- c) If the team cannot identify a suitable response to an identified risk, which risk response strategy would they apply? Justify your answer. 8 (CO4) (PO2)